

Cooper Wooley

coopermw03@gmail.com | (573) 620-3292 | linkedin.com/in/cooperwooley | cooperwooley.github.io

Education

Missouri University of Science and Technology

B.S. Computer Science, Mathematics Minor

Dec 2025

GPA 3.84

Relevant Coursework: Machine Learning, Deep Learning, LLMs, Algorithms, Data Structures, Databases, Operating Systems, Networks, Security, Microcontrollers

Technical Skills

- **Languages:** Python, C/C++, JavaScript, SQL, HTML/CSS
- **Tools/Frameworks:** Linux, Docker, Git, VS Code, Arduino IDE, LaTeX

Experience

Formula Electric Design Team

Software Member

Aug 2023 - Present

- Designed and implemented embedded software for microcontrollers (Teensy, ESP32) using C++ in the Arduino IDE to support the team's electric race car systems
- Extended dashboard class architecture to add object-oriented display control for dual-screen setups, implementing static/dynamic text rendering with hash maps for efficient updates
- Developed an interactive data acquisition (DAQ) visualization interface using JavaScript and Chart.js for real-time telemetry and sensor analytics
- Collaborated to board state logic, auxiliary function libraries, and early sensor testing frameworks to accelerate development for new team members

City of Sikeston - Public Works, Parks and Recreation

Sikeston, MO

Seasonal Skilled Worker

May 2023 - Aug 2025 (Summers)

- Led field maintenance for baseball and soccer setups, ensuring optimal event preparation
- Coordinated daily field marking and equipment setup
- Participated in tornado and flood recovery efforts, safely clearing debris and restoring public spaces

Sikeston Public School District

Sikeston, MO

IT Internship

May 2022 – Aug 2022

- Established internal network infrastructure by configuring distribution frames
- Deployed switches and wireless access points; terminated fiber optic and Cat6 cabling
- Performed hardware troubleshooting and system reimaging

Projects

ParkVision (Capstone Project) – Python, OpenCV, Flask, PostgreSQL, Docker

- Built a smart parking dashboard backend with Flask and PostgreSQL, containerized using Docker for scalability
- Engineered OpenCV-based vision pipelines for automated parking space detection and occupancy analysis
- Designed REST APIs for real-time lot initialization and video-based status retrieval
- Implemented edge detection, reference frame differencing, and occupancy mapping for accurate car tracking

CS2 Workshop Map Queue Bot – Python, Docker, Discord.py, Python-Valve

- Created a Discord bot managing automated 10-player CS2 custom matches with ready checks and map veto logic
- Implemented Dockerized match server orchestration for seamless spin-up, IP delivery, and teardown
- Enabled balanced captain-based team formation with persistent queue management